

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN

Practical I

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1.Environment Setup and JavaScript Basics

a. Install Node.js and Visual Studio Code, and verify installation.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS C:\MERN Practical> node -v
v24.17.0
● PS C:\MERN Practical> npm -v
11.13.0
```

b. Create a simple Node.js file and run it using the terminal.

CODE

```
console.log("Hello, Welcome to 1st Practical of MERN");
console.log("Sneha");
```

OUTPUT

```
● PS C:\MERN Practical> node pract1b.js
Hello, Welcome to 1st Practical of MERN
Sneha
```

c. Write a Node.js script demonstrating basic JavaScript commands (console log, variables, and arithmetic operations).

CODE

```
let num1 = 20;
let num2 = 20;

console.log("First number:", num1);
console.log("Second number:", num2);

console.log("Addition:", num1 + num2);
console.log("Subtraction:", num1 - num2);
```

```
console.log("Multiplication:", num1 * num2);  
console.log("Division:", num1 / num2);
```

OUTPUT

```
PS C:\MERN Practical> node pract1c.js  
First number: 20  
Second number: 20  
Addition: 40  
Substraction: 0  
Multiplication: 400  
Division: 1
```

d. Create and run a simple program using variables and functions.

CODE

```
let c = 10;  
let d = 5;  
  
function add(a, b) {  
    return a + b;  
}  
  
function multiply(a, b) {  
    return a * b;  
}  
  
console.log("First Number: ", c);  
console.log("Second Number: ", d);  
  
console.log("Addition: ", add(c, d));  
console.log("Multiplication: ", multiply(c, d));  
  
console.log("Sneha");
```

OUTPUT

```
PS C:\MERN Practical> node pract1d.js  
First Number: 10  
Second Number: 5  
Addition: 15  
Multiplication: 50  
Sneha
```

e. Demonstrate use of operators and control statements in JavaScript

CODE

```
let a = 10;
let b = 5;

console.log("\nComparison Operators");
console.log("a > b: ", a > b);
console.log("a < b: ", a < b);
console.log("a == b: ", a == b);
console.log("a != b: ", a != b);

console.log("\nIf-else Statement");
if (a>b) {
    console.log("A is greater")
} else {
    console.log("B is greater")
}

console.log("\nSwitch Statement");
let day = 2;
switch (day){
    case 1:
        console.log("Monday");
        break;
    case 2:
        console.log("Tuesday");
        break;
    case 3:
        console.log("Wednesday");
        break;
    default:
        console.log("Invalid input");
}
console.log("Sneha");
```

OUTPUT

```
If-else Statement  
A is greater  
  
Switch Statement  
Tuesday  
Sneha
```

f. Write programs using loops and template literals.

CODE

```
let number = 5;  
  
console.log(`Multiplication Table of ${number}`)  
  
for (let i = 1; i<=10; i++)  
{  
    console.log(`${number} x ${i} = ${number*i}`);  
}  
  
console.log("Sneha");
```

OUTPUT

```
PS C:\MERN Practical> node pract1f.js  
5 x 1 = 5  
5 x 2 = 10  
5 x 3 = 15  
5 x 4 = 20  
5 x 5 = 25  
5 x 6 = 30  
5 x 7 = 35  
5 x 8 = 40  
5 x 9 = 45  
5 x 10 = 50  
Sneha
```